AI-Driven Real-Time Visual Feedback for Neurofeedback

Generating Stimuli with GenAI under Design Guidelines

by

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Abstract

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This project explores the application of generative AI for producing real-time adaptive visual feedback stimuli in neurofeedback applications, specifically targeting alpha wave training for relaxation and tinnitus treatment. We systematically evaluated multiple state-of-the-art generative models (LTX Video, Matrix Game, StreamDiffusion, Stable Diffusion variants) and developed a novel pipeline combining Stable Diffusion XL (SDXL) with vision-language model (VLM) based prompt engineering and Spherical Linear Interpolation (SLERP) in latent space. Our approach addresses the fundamental constraint that current diffusion models cannot generate video in true real-time (24+ fps) while maintaining aesthetic quality and design guideline compliance. By pre-generating keyframes conditioned on training success metrics and interpolating them during real-time playback, we achieve smooth, engaging visual feedback that meets clinical requirements for aesthetic quality, understandability, low cognitive load, fixation stability, universal acceptability, and sustained engagement. Preliminary user testing validates the feasibility of this hybrid approach for clinical neurofeedback applications.

Contents

Nomenclature

Abbreviations

Abbreviation	Definition
NFB	Neuro Feedback
MR	Magnetic Resonance
DiT	Diffusion Transformer
VAE	Variational autoencoder
U-Net	U-shaped convolutional neural network
LTX	Lighttricks
SD	Stable Diffusion
SDXL	Stable Diffusion Xl
SVD	Stable Video Diffusion
ADD	Adversarial Diffusion Distillation
VLM	Vision Language Model
EEG	Electroencephalogram
LoRA	Low-Rank Adaptation
MLP	Multilayer Perceptron
SwiGLU	Combination of activation functions Swish and GLU
RMSNorm	Root Mean Square Layer Normalization
CPU	Central processing unit
GPU	Graphics processing unit
CUDA	Compute Unified Device Architecture
RAM	Random-access memory
LPIPS	Learned Perceptual Image Patch Similarity

1 Introduction

Neurofeedback (NFB) [1, 2] is a non-invasive brain–computer interaction technique in which individuals learn to modulate their neural activity through real-time sensory feedback. Since early demonstrations by Kamiya [3] on alpha-wave discrimination and Sterman [4] on operant conditioning of sensorimotor rhythms, neurofeedback has been explored as a therapeutic and training tool across a range of clinical contexts, including attention disorders, anxiety, chronic pain, and tinnitus.

Tinnitus [5], defined as the perception of sound in the absence of an external source, affects a from 4% to 37% worldwide and from 9% to 29% [6, 7] in Europe proportion of the population and can severely impair quality of life in chronic cases. Existing treatments primarily focus on reducing distress rather than altering the perceptual intensity of tinnitus itself. Neurofeedback has emerged as a promising complementary approach by directly targeting neural correlates associated with tinnitus, particularly through alpha/delta modulation in frontal and fronto-central regions [8].

Beyond the choice of neural targets, the effectiveness of neurofeedback critically depends on how brain activity is translated into perceptible feedback. Prior work indicates that visual feedback characteristics influence learning outcomes, engagement, and self-regulation success [9, 10], yet systematic investigation of visual design in neurofeedback remains limited. This gap has motivated interdisciplinary efforts combining neuroscience, psychology, and design to explore how generative visual systems can support neurofeedback training while remaining perceptually accessible and clinically viable.

Recent advances in generative models—especially diffusion-based image synthesis and vision–language models—offer new possibilities for adaptive, expressive visual feedback that evolves continuously with brain state [11–13].. At the same time, these technologies introduce practical and conceptual challenges related to controllability, temporal coherence, and alignment with clinical constraints [14, 15]. This project situates itself at the intersection of neurofeedback research and generative AI, investigating how modern generative pipelines can be leveraged to produce visually coherent, signal-driven feedback while respecting the constraints of neurofeedback practice.

2
Approach

2.1. Objectives and Research Hypotheses

This project investigates the design of visually rich yet cognitively lightweight NFB stimuli under strict real-time and clinical constraints. Recent research in generative artificial intelligence has demonstrated that AI-generated visual content can surpass human-made imagery in engagement, attention capture, and aesthetic quality [16, 17], suggesting potential applications beyond traditional marketing contexts. The initial working hypothesis is that *generative artificial intelligence*, when combined with appropriate design principles and established neurofeedback guidelines, can produce adaptive visual feedback that is more engaging and perceptually suitable than traditional approaches. Current neurofeedback systems typically rely on simple visual animations or parameter-driven displays [18], which may contribute to monotony during extended training sessions. In contrast, this work proposes that generative models can create varied, aesthetically rich stimuli while still adhering to core neurofeedback requirements—including low cognitive load, gaze stability, and temporal smoothness [19]—which are essential for effective brain training and clinical acceptability.

The work follows an exploratory approach, beginning with a systematic examination of existing generative image and video models. Through this exploration, additional questions emerge regarding the feasibility of real-time generation and the mechanisms by which continuous neural signals can be mapped to controlled visual evolution.
As a result, a second hypothesis is formulated: *true real-time video generation* using current generative models is not technically feasible at the frame rates and quality required for clinical neurofeedback, motivating alternative strategies that decouple generation from time-critical presentation. In parallel, a third hypothesis emerges concerning *prompt engineering*: manual prompt design proves difficult to scale and standardize, suggesting that vision–language models can automate prompt construction and evolution in a more consistent and reproducible manner.
To investigate these hypotheses, the project proceeds iteratively, as illustrated on the figure 2.1.

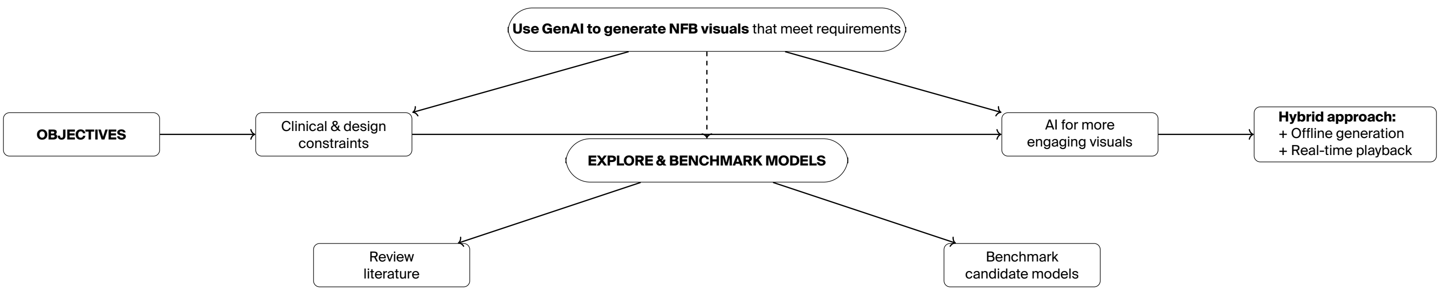

Figure 2.1: Conceptual overview of the project objectives, exploratory phase, and resulting hybrid generation strategy for neurofeedback visualization.

2.2. Related Work

This section reviews prior work relevant to the proposed approach, with a particular focus on generative models capable of producing adaptive visual stimuli. Beyond summarizing existing systems, the objective is to identify whether recent advances in fast image and video generation can satisfy the specific constraints of neurofeedback visualization: continuous temporal coherence, fine-grained controllability, low cognitive load, and compatibility with signal-driven modulation. Rather than assuming a priori that generative video models are suitable, this work adopts an exploratory perspective. Several classes of modern generative pipelines are examined and experimentally evaluated to assess whether they can support visually coherent, abstract feedback evolving smoothly as a function of a continuous neural signal.

2.2.1. Neurofeedback Systems

Fundamentals Neurofeedback (EEG biofeedback) is based on operant conditioning, enabling individuals to learn neural self-regulation through real-time sensory feedback of their brain activity. Electrical signals measured via EEG are typically analyzed in canonical frequency bands, ranging from slow delta rhythms (deep sleep) to fast gamma activity (high-level cognition). Among these, alpha-band activity (8–12 Hz), associated with relaxed wakefulness, is the most extensively validated target for neurofeedback, particularly in relaxation and stress reduction. Meta-analytic evidence shows that alpha neurofeedback outperforms SMR-based protocols for neural modulation, with statistically significant effect sizes across feedback modalities [20, 21]. Moreover, interventions using complex or audiovisual stimuli demonstrate larger training effects compared to unimodal feedback [20].

Tinnitus-Oriented Neurofeedback In the context of tinnitus, Güntensperger et al. (2019) [22] demonstrated that neurofeedback protocols rewarding individualized alpha frequencies lead to significant and durable reductions in tinnitus distress when administered in repeated 20-minute sessions. These protocols typically target frontal–central electrode sites and combine alpha enhancement with delta inhibition. The therapeutic rationale is grounded in maladaptive plasticity models of tinnitus, where increased frontal alpha power is thought to support top-down inhibition of aberrant auditory activity. Despite promising outcomes, the translation of these protocols into clinical practice remains limited, and the influence of feedback design on treatment efficacy is still insufficiently understood.

2.2.2. Design Requirements for Neurofeedback

Cognitive Load and Learnability Effective neurofeedback (NFB) design must balance engagement with cognitive demands to avoid disrupting neural self-regulation [23]. Cognitive load can be decomposed into: (i) intrinsic load, inherent to the self-regulation task; (ii) extraneous load, induced by poorly designed feedback; and (iii) germane load, supporting learning. Excessive extraneous load from complex or ambiguous visuals impairs learning and may explain why between 16% and 57% of users fail to achieve self-regulation [24, 25], with most studies reporting non-responder rates around 30–50% [25]. Optimal feedback therefore minimizes extraneous load through intuitive visuals while maintaining sufficient germane load via meaningful, variable reinforcement.

Visual Attention and EEG Constraints Eye movements introduce strong EEG artifacts, particularly at frontal sites, and can be misinterpreted as genuine neural changes [26]. Visual design must therefore discourage gaze shifts and saccades. Key requirements include central fixation, minimal peripheral complexity, smooth and gradual transitions, and comfortable luminance levels. These constraints support both signal integrity and the focused, meditative state associated with alpha enhancement.

Aesthetics, Engagement, and Familiarity Aesthetic quality directly affects user engagement over repeated training sessions. Prior findings show a preference for circular or organic forms, centered compositions, gradual color transitions, and abstract representations, which reduce semantic interpretation demands. The principle of *Super Normality*—using familiar, universally recognizable visual cues—facilitates comprehension and lowers novelty-induced cognitive load, while controlled variation is required to prevent habituation during extended sessions.

Gestalt Principles and Visual Organization Gestalt principles provide a framework for managing visual complexity while preserving clarity. Proximity and similarity support perceptual grouping, continuity favors smooth evolving patterns, closure allows completion of incomplete forms, and clear figure–ground separation enhances legibility. Applying these principles ensures aesthetic coherence and low organizational complexity as feedback dynamically evolves.

Design Guidelines from Iterative Studies Iterative user studies conducted at the EPFL+ECAL Lab yielded the following guidelines: (i) spatial encoding of performance (e.g., size, spread, density) is the most easily understood message variable; (ii) circular, centrally focused elements align with both perceptual and technical constraints; (iii) familiar visual metaphors improve accessibility; and (iv) systematic use of Gestalt principles maintains visual clarity. Three feedback dynamics were developed: *static* (accumulative circular patterns), *kinetic* (continuous brush-like motion encoding performance via speed and thickness), and *organic* (reaction–diffusion systems with continuously evolving forms). Each was combined with three evolution modes: none, continuous, or discrete morphological transitions.

Final Design Objectives The final objectives are to ensure that feedback is aesthetically pleasing, easily understandable, cognitively lightweight, resistant to eye drifting, universally acceptable, and sufficiently varied to avoid boredom.

2.2.3. Fast Generative Models

To investigate whether contemporary generative systems can meet neurofeedback constraints, we evaluate several classes of fast image and video generation models. These models were selected to represent complementary trade-offs between speed, temporal coherence, controllability, and expressive capacity.

Specifically, we examine: (i) video diffusion models designed to synthesize short coherent clips, (ii) real-time world simulation models capable of frame-rate generation, (iii) pipeline-level accelerations of diffusion for near–real-time image synthesis, and (iv) high-quality image diffusion models repurposed for sequential generation.

Each model class is analyzed both architecturally and experimentally to assess whether it can support smooth, signal-driven visual evolution.

Video Diffusion Models

LTX Video LTX Video [27] is a diffusion-based video generation model built on a Diffusion Transformer (DiT) backbone. It combines a highly compressed Video-VAE—using spatiotemporal downsampling with a compression ratio of 1:192—with separate spatial and temporal attention mechanisms and rotary positional embeddings. This architecture enables faster-than-real-time video synthesis (5 seconds of 24 fps video in 2 seconds on an H100 GPU), making it a strong candidate for real-time feedback scenarios.
LTX Video is selected for evaluation because neurofeedback stimuli are inherently temporal and typically presented as continuous visual sequences. Its ability to generate coherent video at high speed while maintaining temporal consistency makes it suitable for investigating continuous signal-driven visual feedback.

Stable Video Diffusion Stable Video Diffusion [28] extends latent diffusion models with temporal convolutions and attention layers to generate short video clips (14-25 frames) from a single input image. The model uses a 1.5B parameter architecture with 656M parameters devoted to temporal processing. While temporally coherent over brief durations, SVD does not support text-based conditioning and relies on auxiliary motion parameters such as motion buckets and noise augmentation. Clip duration is limited to approximately 4 seconds, and motion frequently manifests as camera-like effects rather than semantic evolution.
SVD is included because it represents a direct attempt to adapt image diffusion to video synthesis, allowing us to investigate whether img2vid approaches can bridge the gap between static and dynamic neurofeedback stimuli.

Image Diffusion Models for Sequential Generation

Stable Diffusion and SDXL Stable Diffusion [11] introduced latent diffusion models that operate in a compressed latent space, enabling high-resolution synthesis with reduced computational cost. Stable Diffusion XL (SDXL) [12] extends this architecture with a substantially larger U-Net (2.6B parameters, three times larger than previous versions), higher-resolution latent representations, and dual text encoders (OpenCLIP ViT-bigG and CLIP ViT-L), resulting in improved semantic fidelity and compositional control.

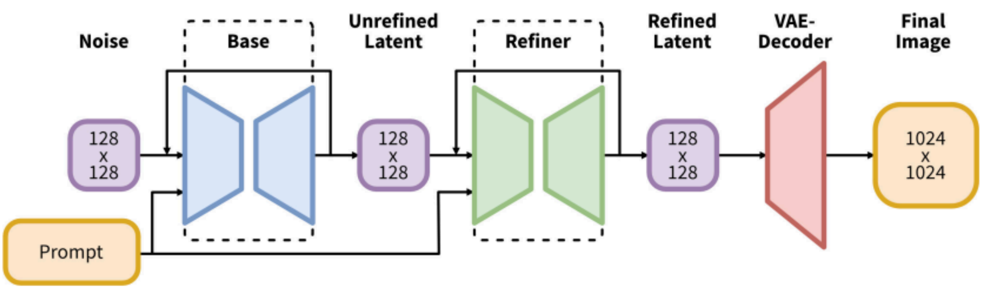

Figure 2.2: SDXL architecture. In a first step, the base model is used to generate (noisy) latents, which are then further processed with a refinement model specialized for the final denoising steps.

Although SDXL is not a video model, it was selected because its `img2img` formulation enables deterministic, incremental frame-to-frame evolution. Each frame can be generated by partially noising the previous one and re-denoising it under controlled textual guidance. Parameters such as denoising strength, guidance scale, and negative prompts provide explicit and interpretable control channels that can be mapped to a continuous signal, making it particularly relevant for investigating parametric control in neurofeedback systems.

World Simulation Models

Matrix-Game Models Neural world simulation models such as Matrix-Game 2.0 [29] are trained on large-scale gameplay datasets (approximately 1,200 hours of annotated interactive video) and predict future frames conditioned on the current state and discrete player actions. These models achieve real-time performance (25 fps) through few-step autoregressive diffusion and exhibit strong temporal consistency due to their deterministic simulation dynamics.
They are selected for exploration because they appear, at first glance, to satisfy the real-time requirement of neurofeedback. We investigate whether discrete control inputs (e.g., keyboard or mouse actions, see Figure 2.3) could be reinterpreted as proxies for neural signals, potentially enabling game-like interactive neurofeedback environments.

Pipeline-Level Acceleration

StreamDiffusion and SDXL-Turbo StreamDiffusion [30] accelerates diffusion-based image synthesis by optimizing the sampling pipeline rather than modifying the model architecture. Techniques such as stream batch processing, residual classifier-free guidance (RCFG), and hardware-specific optimizations enable frame rates exceeding 90 fps on an RTX 4090, achieving up to 59.56× speedup over standard pipelines.

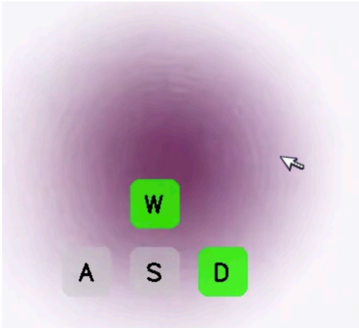

Figure 2.3: Interactive simulation, controllable with keyboard and mouse

SDXL-Turbo [31] employs Adversarial Diffusion Distillation (ADD) to enable single-step image generation, combining score distillation with adversarial training to maintain image quality while reducing sampling steps from 50 to 1-4. This approach generates 512×512 images in 207ms on an A100 GPU.

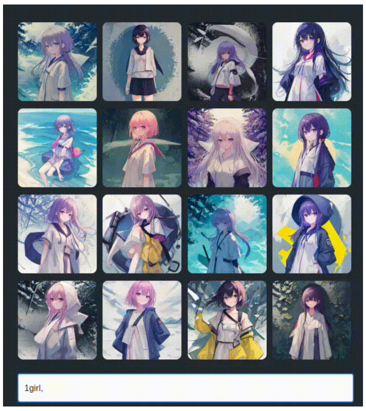

Figure 2.4: Interactive txt2img demo of StreamDiffusion pipeline

StreamDiffusion and SDXL-Turbo are selected because they closely resemble true real-time generation (see Figure 2.4), making them ideal candidates for investigating whether pipeline-level optimizations alone can satisfy neurofeedback timing constraints without sacrificing visual quality.

LoRA Fine-Tuning for Style Control

Low-Rank Adaptation Low-Rank Adaptation (LoRA) [32] enables parameter-efficient fine-tuning by injecting low-rank updates into selected layers of a frozen diffusion model. LoRA reduces the number of trainable parameters by up to 10,000× while maintaining model quality, making it practical to create multiple stylistic variants without full retraining. In principle, LoRA could enforce stylistic consistency across frames while allowing rapid switching between different visual aesthetics during neurofeedback sessions.

2.2.4. Vision--Language Models for Prompt Engineering

Qwen2.5-VL-7B-Instruct Qwen2.5-VL-7B-Instruct [33] is a large-scale vision–language model designed for high-fidelity multimodal understanding and instruction following. The model comprises approximately 8B parameters and integrates a native vision encoder with a 7.4B-parameter Qwen2.5 language backbone [34].
The vision encoder (≈600M parameters) is a dynamic-resolution Vision Transformer that processes images at variable token budgets, using windowed attention to maintain linear computational scaling. Spatial structure is encoded via 2D Rotary Position Embeddings (M-RoPE), enabling precise localization and layout reasoning. Visual representations are projected into the language space through an MLP-based merger augmented with multimodal rotary embeddings, allowing unified spatial–semantic reasoning. The language backbone employs grouped-query attention, SwiGLU activations, and RMSNorm, yielding strong instruction-following performance with efficient inference.
These architectural choices allow Qwen2.5-VL to adapt computation to image complexity and operate natively across resolutions without traditional normalization techniques. The model was trained on a curated dataset expanding from 1.2 trillion to 4.1 trillion tokens, encompassing diverse multimodal data types including documents, videos, and interactive applications. This makes it particularly effective for detailed visual analysis rather than coarse captioning.

Capabilities Relevant to Prompt Engineering On standard vision–language benchmarks, Qwen2.5-VL achieves state-of-the-art or near state-of-the-art performance. The 7B model outperforms comparable open-source models on key benchmarks including MathVista [35] (68.2% accuracy for mathematical reasoning in visual contexts), DocVQA [36]

(95.7% accuracy for document understanding), and MMMU (58.6% on massive multi-discipline multimodal understanding).

For the purposes of generative prompt engineering, its most relevant strengths lie in fine-grained visual decomposition and structured reasoning. The model reliably extracts geometric structure, color palettes, compositional balance, and stylistic attributes. Crucially, Qwen2.5-VL supports structured output generation—it can produce valid JSON outputs for coordinates, object attributes, and hierarchical scene descriptions with high consistency. This capability enables systematic downstream use in generative pipelines rather than ad hoc natural language prompting, making it particularly suitable for automated prompt engineering workflows where machine-readable descriptions must be generated from visual stimuli.

2.2.5. Limitations of Existing Approaches and Retained Capabilities

The reviewed literature reveals that no existing generative model natively satisfies the combined requirements of neurofeedback visualization: continuous signal responsiveness, temporal coherence, abstract expressiveness, and sub-second latency. Current systems optimize for a single dimension—speed, realism, or stability—at the expense of others. Video diffusion models offer temporal coherence but condition at the clip level and bias toward photorealism, limiting abstract representation. Real-time image pipelines enable fine-grained control but lack temporal priors, causing frame-to-frame instability. World simulation models maintain consistency through learned dynamics but remain constrained to discrete controls and predefined environments. LoRA-based style adaptation requires large homogeneous datasets rarely available in clinical neurofeedback contexts.

However, diffusion-based image models retain critical capabilities: flexible prompt conditioning, support for abstract visual semantics, and explicit parametric control. These properties, combined with insights from video modeling and pipeline optimization, motivate a hybrid architecture that decouples high-quality generation from real-time playback.

The approach proposed in this work leverages these retained capabilities while addressing identified gaps through offline pre-generation and signal-aligned temporal interpolation, enabling continuous visual feedback that is both aesthetically expressive and computationally feasible.

3
Experiments

3.1. Hardware and Software Environment

3.1.1. Hardware

All experiments were conducted on a single high-end workstation. The configuration is summarized in Table 3.1.

Table 3.1: Hardware configuration

Operating system	Ubuntu 24.04.3 LTS (x86_64)
CPU	Intel Core i9-14900K (24 cores, 32 threads, up to 6.0 GHz)
GPU	NVIDIA GeForce RTX 4090 (24 GB VRAM)
CUDA	CUDA Toolkit 13.0 (nvcc 13.0.88, driver 580.95.05)
RAM	128 GB DDR5 (125 GiB total, 119 GiB available)
Storage	NVMe SSD (215 GB system partition, 124 GB available; 1 TB device total)

The RTX 4090 provided sufficient compute and memory capacity for SDXL image generation (4–6 s per 1024×1024 image at 40–50 denoising steps) and near–real-time VAE decoding (50–100 ms per frame). The project's software environment, already installed on the system, occupying approximately 6 GB of disk space, did not constrain GPU memory usage: the available 24 GB of VRAM was sufficient to simultaneously load SDXL, dual text encoders, the VAE, and the Qwen2.5-VL model using reduced-precision weights. All models were loaded using mixed-precision inference (fp16/int8), with attention and VAE modules optimized for inference. Details of the software environment are described in the following subsection.

3.1.2. Software Stack

The software stack was based on Python 3.11 and PyTorch 2.9 with CUDA 13.0 support. Diffusion pipelines were implemented using `diffusers`, while text and vision–language components relied on `transformers` and `accelerate`. Image and video processing used standard libraries including Pillow, OpenCV, and FFmpeg, with numerical computation and visualization handled by NumPy and Matplotlib. Experiments were developed and tested using Jupyter and Python scripts.

3.1.3. Model Checkpoints

The experiments relied on publicly available pretrained checkpoints listed in Table 3.2.

Table 3.2: Model checkpoints used in the experiments

Model	**Checkpoint**	**Precision**
SDXL Base	`stabilityai/stable-diffusion-xl-base-1.0`	FP16
Qwen2.5-VL	`Qwen/Qwen2.5-VL-7B-Instruct`	BF16
SDXL VAE	Included in base model	FP16
CLIP ViT-L/14	Included in SDXL	FP16
OpenCLIP ViT-bigG/14	Included in SDXL	FP16

All checkpoints were used under their respective open licenses and loaded in reduced precision to balance memory usage and inference speed.

3.2. Evaluation of Models

We evaluate five generative models and pipelines for image-conditioned visual sequence generation: Stable Video Diffusion (SVD), LTX-Video, Stable Diffusion XL (SDXL), Matrix-Game, and StreamDiffusion. For each method, we generate a sequence of 24 frames at 512×512 resolution conditioned on a single initial input image and a fixed textual prompt. Image-based models (SDXL and StreamDiffusion) are run in an iterative image-to-image setting, where the output of frame t is used as the conditioning input for frame $t+1$ to induce temporal evolution, while video-native models (SVD, LTX-Video, Matrix-Game) directly produce a multi-frame sequence in a single forward pass. All methods are evaluated under comparable resolution and frame count constraints, with fixed random seeds to ensure reproducibility.

Performance Metrics For each model, effective throughput was measured as the number of generated frames divided by the total wall-clock generation time, reported as effective frames per second (fps). GPU memory usage was measured using the maximum reserved VRAM reported by the deep learning framework during generation. Reported values correspond to peak usage observed during the task.
Temporal coherence was quantified using the Learned Perceptual Image Patch Similarity (LPIPS) [37] metric computed between consecutive frames. For each sequence, LPIPS values were calculated frame-to-frame, yielding a distribution from which the mean and maximum were extracted. These statistics capture both overall smoothness and the presence of abrupt perceptual changes. The results are summarizes on table 4.1.

Derived Ordinal Scores In addition to raw performance measurements, models were assigned ordinal scores along six qualitative dimensions relevant to neurofeedback visualization, summarized in Table 4.2. These scores were derived from explicit, predefined criteria rather than subjective judgment.
Speed scores were assigned based on effective fps ranges, with thresholds chosen to reflect practical real-time constraints. Temporal coherence scores were derived directly from LPIPS statistics, using predefined ranges of mean LPIPS values. Quality scores were computed from a composite, normalized metric aggregating several frame-level proxies: image sharpness (Laplacian variance), background purity (near-white pixel ratio), background texture energy (high-frequency content), and centeredness of the foreground. Each component was normalized across models before aggregation, and models were ranked by percentile.
Control scores reflect the number and type of native conditioning interfaces exposed by each system, including text prompts, negative prompts, guidance or strength parameters, and continuous controllable variables that induce predictable visual changes. Training fit scores were assigned by evaluating alignment between the model's training distribution and abstract neurofeedback design requirements, based on the presence or absence of specific criteria such as support for non-photoreal visuals and freedom from fixed environments.
Finally, conditioning feasibility scores capture whether a continuous scalar signal $s \in [0, 1]$ can be mapped to a smooth, predictable, and monotonic visual evolution without retraining. This score integrates the existence of controllable parameters, the ability to vary them densely over time, empirical monotonicity of selected visual metrics, and local temporal smoothness as measured by LPIPS.

3.3. Adopted Strategy: Offline Generation with Real-Time Playback

Following the comparative evaluation of candidate models, we adopted a hybrid strategy that separates computationally intensive generation from time-critical presentation. This design directly addresses the inability of current diffusion models to generate high-quality video at real-time frame rates, while still enabling adaptive feedback during neurofeedback sessions.

3.3.1. System Architecture

The pipeline is divided into two clearly separated phases:

Offline Phase (Pre-generation) A complete set of visual states is generated ahead of time:

1. **Reference Image Selection:** A representative image defining the target aesthetic is chosen.
2. **Automated Prompt Engineering:** A vision–language model (VLM) analyzes the reference and produces a sequence of progressively evolving prompts.
3. **Keyframe Generation:** SDXL generates a fixed number of keyframes (20 in our configuration) using the `img2img` pipeline.
4. **Latent Interpolation:** Intermediate frames are synthesized between keyframes using SLERP in latent space.
5. **Frame Indexing:** All decoded frames are stored with an associated normalized signal value.

Online Phase (Playback) During neurofeedback:

1. A continuous neural feature derived from EEG, corresponding to alpha-band activity (8–12 Hz), is received from the signal processing pipeline.
2. The signal is normalized to a bounded control range and mapped to a pre-generated frame index.
3. The corresponding frame is loaded and displayed at the target frame rate (16–24 fps).

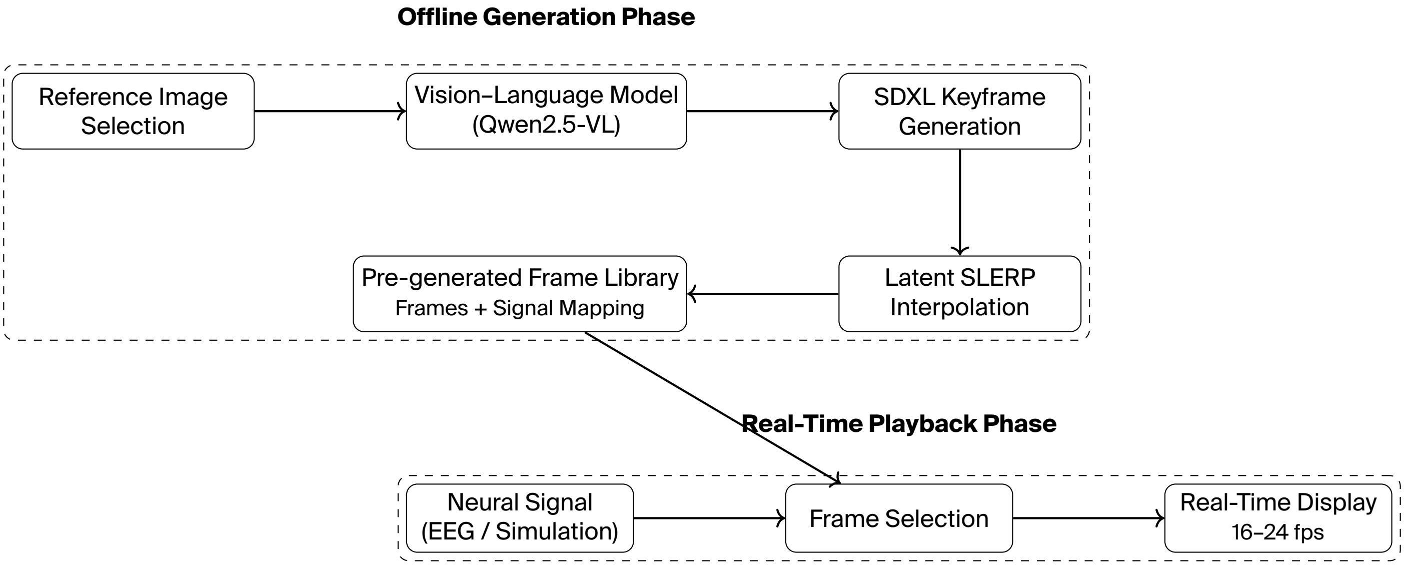

Figure 3.1: Compact overview of the hybrid neurofeedback visualization pipeline. Offline generation produces a library of frames indexed by signal value, enabling low-latency real-time playback without online image synthesis.

This architecture enables responsive feedback with minimal latency, while ensuring full control over visual quality and aesthetics. Figure 3.1 provides an overview of the resulting pipeline, from reference image selection and prompt generation to offline synthesis and signal-driven playback.
A deterministic, monotonic mapping links normalized signal values to discrete frame indices:

$$\text{frame\_index} = \text{round}\big(\text{signal} \times (N-1)\big),$$

where N is the total number of frames. With $N = 721$, the signal resolution is approximately 1.4×10^{-3} per frame. This mapping ensures reproducible visual states for identical signal values and simplifies integration with neurofeedback systems.

3.4. Conditioning Mechanisms

We iteratively developed conditioning mechanisms in four stages, progressively increasing control, consistency, and scalability.

3.4.1. Stage 1: No Prompt

Generating images without text conditioning resulted in highly variable outputs with no reliable control over aesthetics or structure. This approach was unsuitable for design-guideline compliance.

3.4.2. Stage 2: Single Manual Prompt

A single, carefully crafted prompt produced visually consistent results and served as a strong baseline. However, static prompts cannot express gradual evolution and require substantial manual tuning.

3.4.3. Stage 3: Manual Prompt Sequences

Sequences of hand-written prompts enabled explicit control over visual progression. While effective, this approach was time-consuming, stylistically inconsistent, and inherently subjective, limiting scalability.

3.4.4. Stage 4: Automated VLM-Based Reverse Engineering (Final Approach)

The final system uses Qwen2.5-VL-7B-Instruct to automate prompt sequence generation:

1. **Image Analysis:** The VLM extracts key visual attributes (composition, palette, style) from a reference image.
2. **Evolution Expansion:** A brief user-defined goal is expanded into a detailed visual specification under explicit style constraints.
3. **Prompt Sequence Generation:** The VLM produces a sequence of prompts mapped to evenly spaced signal values.

This approach provided consistent phrasing, richer descriptions, reduced manual effort, and objective interpolation across signal ranges.

3.5. Parameter Selection

Key parameters were tuned empirically; selected values are summarized in Table 3.3. These settings provided a balance between smooth visual evolution, generation time, and prompt adherence.

Parameter	Values Tested	Selected Value	Rationale
num_prompts	7, 13, 20, 30	21	Balance between smoothness and generation time
analysis_type	detailed, brief, technical	brief	Sufficient detail without excessive length
num_interpolations	3, 5, 10, 35, 50	35	Smooth motion without excessive decode time
strength (SDXL img2img)	0.15, 0.25, 0.35, 0.45, 0.6	0.45	Visible evolution without discontinuity
guidance_scale	7.5, 10, 15, 20	15	Strong prompt adherence without over-saturation
num_inference_steps	20, 40, 50, 75, 100	50	Quality vs. speed tradeoff

Table 3.3: Parameters tuning for SDXL pipeline

The pipeline was implemented as a set of modular scripts for image generation and video assembly. While effective, limitations remain: prompt verbosity growth, occasional stylistic drift or hallucinations, and the absence of learned style adapters (LoRA). These issues were mitigated through explicit style constraints but remain areas for future improvement.

4
Results

4.1. Performance Metrics

4.1.1. Model Comparison

All candidate models were evaluated under a unified benchmarking protocol on the same workstation (see Section 3.2), using a standardized task consisting of the generation of a 24-frame sequence at 512×512 resolution. Depending on the model, sequences were obtained either as native video outputs, streamed image sequences, simulator rollouts, or chained image generations. Each benchmark was repeated multiple times to estimate mean values and variability.

Table 4.1: Benchmark on a standardized 24-frame generation task (512×512).

System	Output	Task	Eff. fps (frames/sec)	Peak VRAM reserved (GB)	LPIPS mean ↓	LPIPS max ↓
LTX Video	video	24 frames	6.44 ± 0.01	17.5 ± 0.0	0.0027 ± 0.002	0.0156 ± 0.0213
SVD	video	24 frames	0.96 ± 0.001	10.16 ± 0.0	0.23 ± 0.13	0.55 ± 0.26
Matrix-Game 2.0	simulator	24 frames	0.81 ± 0.07	16.69 ± 0.0	0.048 ± 0.04	0.21 ± 0.24
StreamDiffusion (sd-turbo)	streamed img	24 frames	13.53 ± 1.09	2.71 ± 0.0	0.032 ± 0.003	0.10 ± 0.02
SDXL (img2img chain)	images	24 frames	3.03 ± 0.004	7.69	0.058 ± 0.008	0.095 ± 0.02

Table 4.2: Comparison of evaluated generative models for neurofeedback visualization (ordinal scores)

Model	Speed	Quality	Control	Training Fit	Temporal Coherence	Conditioning Feasibility
LTX Video	2	3	2	1	3	2
SVD	0	1	0	1	0	1
Matrix-Game 2.0	0	0	0	0	2	0
StreamDiffusion (sd-turbo)	3	0	2	2	2	2
SDXL (img2img)	2	2	2	2-3	2	3

Scale: 0 = not supported, 1 = limited, 2 = moderate, 3 = strong.

4.1.2. Qualitative Comparison

Model Comparison from Identical Initial Conditions Figure 4.1 presents a qualitative comparison between *StreamDiffusion*, *LTX Video*, and *SDXL* under matched initial conditions. For each model, generation starts from the same reference image and is guided by an identical textual prompt. Five prompts were selected from existing abstract visual material developed within the ANT project at the EPFL+ECAL Lab, covering a range of minimalist and organic design motifs. Each row corresponds to one prompt, while columns show the resulting output produced by each model. This setup allows direct visual comparison across models when both image conditioning and text conditioning are available.

Matrix-Game Video Evolution Figure 4.2 illustrates the temporal evolution of a video generated by Matrix-Game 2.0. A sequence of evenly spaced frames is extracted from a single generated clip and arranged in temporal order. The figure visualizes how the system evolves its internal state over time when driven by its native simulation dynamics. The displayed frames highlight the characteristic transition toward visually realistic environments, reflecting the model's training on large-scale interactive game datasets.

Stable Video Diffusion (SVD) Evolution Figure 4.3 shows the temporal evolution of a video produced by Stable Video Diffusion (SVD) from a single conditioning image. Frames are sampled uniformly over the generated clip and displayed sequentially. The figure documents how visual structure changes across time under the model's image-to-video generation process, illustrating the nature of temporal variation produced by SVD in the absence of explicit textual or parametric control over intermediate states.

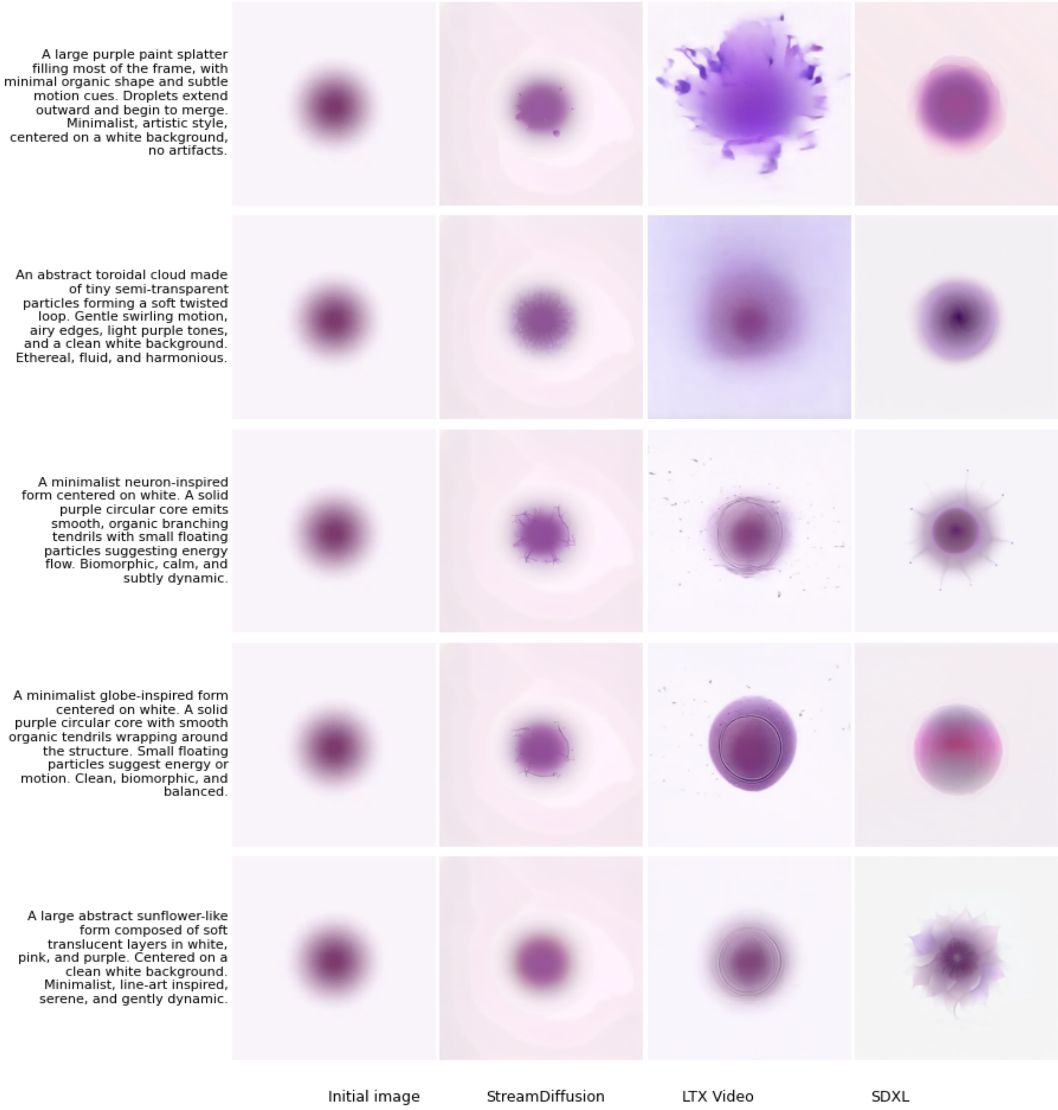

Figure 4.1: Qualitative comparison of SDXL with LTX-Video and Streamdiffusion with turbo-sd. Seed 0 was uniformly applied across all models in which such a parameter could be designated.

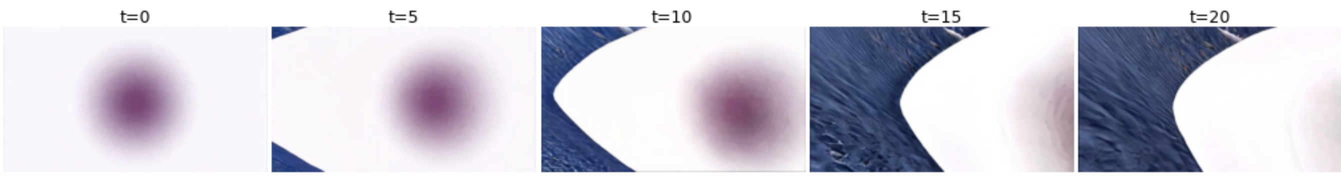

Figure 4.2: Evolution of a video generated by Matrix-Game

4.1.3. Interpolation Quality

Interpolation quality was evaluated by comparing linear interpolation (LERP), spherical linear interpolation (SLERP) [38], and cosine interpolation in latent space. SLERP preserves vector magnitudes during interpolation, which is particularly important for high-dimensional latent spaces where linear interpolation can produce out-of-distribution samples [39]. Quantitative evaluation used the LPIPS metric and norm deviation. LPIPS measures perceptual distance using deep features from pretrained networks, providing better alignment with human perception than traditional metrics like PSNR or SSIM. Table 4.3 summarizes the results.

Table 4.3: Interpolation quality comparison

Method	LPIPS (mean)	LPIPS (max)	Norm Deviation
LERP	0.002	0.009	2%
Cosine	0.003	0.014	1.5%
SLERP	0.002	0.009	0.008%

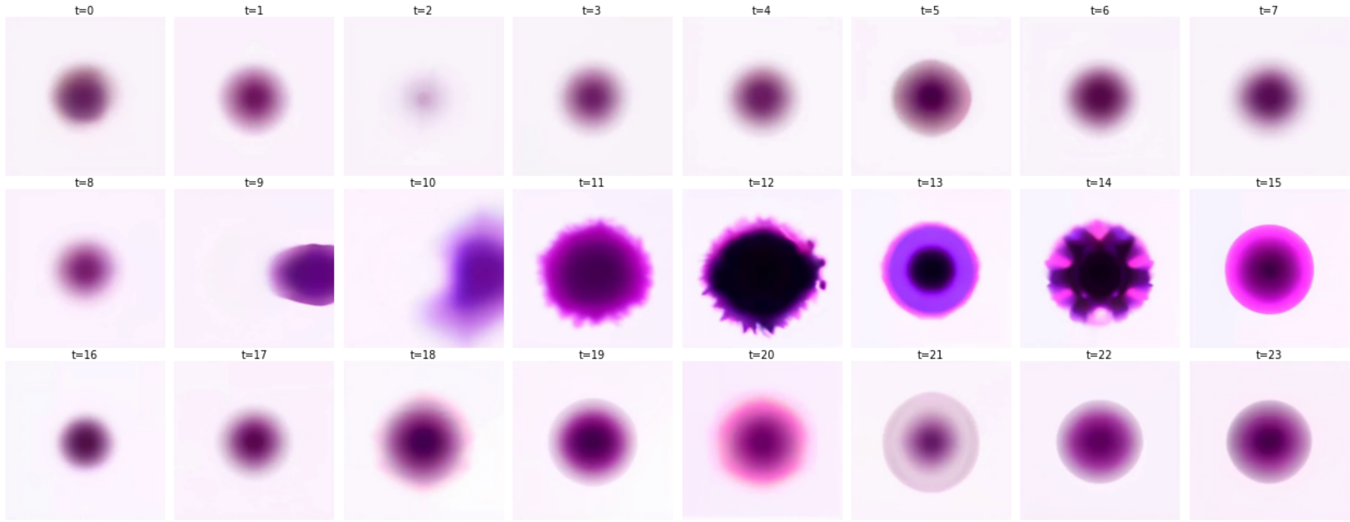

Figure 4.3: Evolution of a video generated by SVD

SLERP achieved the lowest norm deviation (0.008%), indicating superior magnitude preservation compared to LERP (2%) and cosine interpolation (1.5%). All methods produced comparable perceptual quality with LPIPS scores below 0.003 on average, suggesting smooth transitions between latent vectors.

4.1.4. SDXL Pipeline Performance

Offline Generation Table 4.4 summarizes computational requirements during offline preparation of a 30-second sequence at 24 fps (721 frames). Keyframe generation performance was measured for the selected SDXL configuration (512×512, 40–50 steps, guidance scale 15, strength 0.45). The results are computed on 10 identical runs with different seeds.

Table 4.4: SDXL generation and interpolation performance

Stage	Metric	Value (s)
Qwen2.5-VL prompt generation	Mean total generation time	$67. \pm 12.1$
Single keyframe	Mean generation time	1.15 ± 0.01
21 keyframes	Mean total generation time	24.1 ± 0.6
SLERP interpolation	Mean total interpolation time	87.2 ± 3.2
VAE encoding	Mean time per frame	0.03 ± 0.001
VAE decoding	Mean time per frame	0.05 ± 0.0001
Full sequence (721frames)	Mean Total offline time	172.4 ± 15.3

Real-Time Playback Table 4.5 reports resource usage during stimulus playback.

Table 4.5: Real-time playback performance (estimated on a typical modern PC with NVMe SSD; no GPU required).

Component	CPU (%)	RAM (MB)	Latency (ms)
Signal acquisition (simulated)	< 1	≈0	< 0.05
Frame index lookup	< 1	≈0	< 0.01
Frame loading (NVMe SSD)	3–8	200–400	2–6
Display refresh (60 Hz)	1–3	≈0	16.7
Total	5–12	200–400	**19–23**

4.1.5. Storage Requirements

For a single 30-second sequence (685 frames):

- PNG frames (512×512): ≈ 70–170 MB (typically ∼100 MB for minimalist visuals with white backgrounds)
- Prompt and metadata files: < 0.5 MB
- Encoded video (H.264): ≈ 8–15 MB

Total storage per sequence is approximately ∼110–120 MB. A 1 TB NVMe SSD can store on the order of $\sim 8{,}000$–$9{,}000$ unique sequences, corresponding to roughly 65–75 hours of non-repeating stimuli (at 30 seconds per sequence).

4.2. User Testing

4.2.1. Study Design

A preliminary user study was conducted to assess the perceptual suitability of the generated visual stimuli for neurofeedback applications. A total of 18 adult participants (18–65 years) took part in the study.
Participants viewed a set of seven video sequences, each with a duration of one minute. The stimuli represented different visual metaphors (e.g., flow, growth, densification, organic expansion) and were driven by simulated control signals (slow sinusoidal oscillation) to emulate continuous neurofeedback modulation. No EEG data were recorded at this stage; the study focused exclusively on perceptual and experiential aspects of the visuals.
Each session consisted of a short pre-questionnaire (demographics, prior neurofeedback exposure, visual design expertise), followed by stimulus presentation and a post-stimulus evaluation.

Participant Demographics and Expertise Participant demographics (age, gender, education level, employment status) were summarized in Figure 4.4. Self-assessed expertise in visual design and neurofeedback summarized in Figure 4.5.

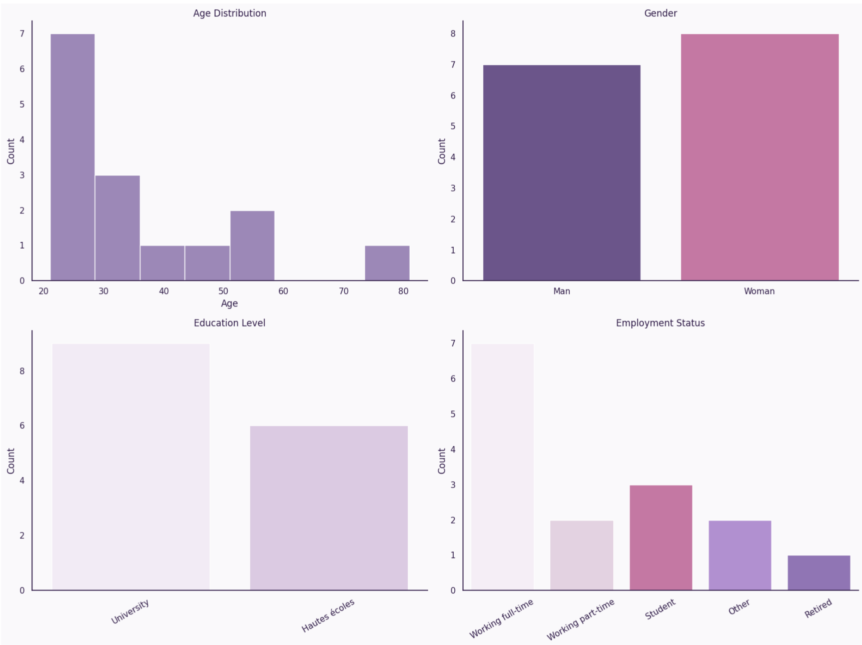

Figure 4.4: Participant demographics: age distribution, gender, education level, and employment status.

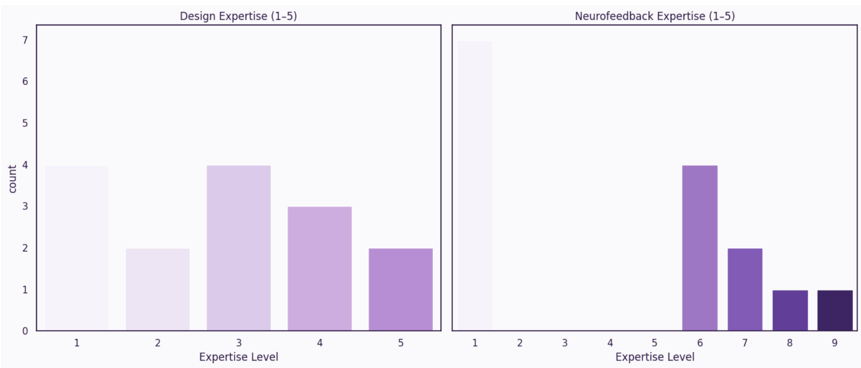

Figure 4.5: Self-reported expertise in visual design and neurofeedback training.

Evaluation Measures To evaluate how different visual stimuli were perceived with respect to neurofeedback design objectives, we collected self-reported subjective ratings across several perceptual and cognitive dimensions identified in prior work on neurofeedback visualization, media perception, and human–computer interaction. The selected measures capture both theoretical constructs—such as abstraction, aesthetic judgment, and engagement—and practical constraints specific to neurofeedback, including interpretability of the signal–visual mapping and cognitive load.
Participants evaluated each stimulus using Likert or semantic differential scales with explicitly defined anchors. Scale ranges were chosen to balance sensitivity and ease of use while remaining compatible with established neurofeedback evaluation practices. The evaluated dimensions were designed to assess whether the intended visual properties were perceived as expected and to characterize how different designs were experienced during sustained observation.

- **Understandability** (5-point Likert scale): assessed how clearly participants could perceive the relationship between changes in the underlying signal and the resulting visual evolution. The scale ranged from *"Extremely difficult"* (1) to *"Extremely easy"* (5), targeting message clarity and learnability.
- **Visual complexity** (6-point semantic differential): measured perceived simplicity versus intricacy of the stimulus, using the bipolar anchors *"Very simple"* (1) and *"Very complex"* (6). This dimension reflects processing fluency and extraneous cognitive load.
- **Abstractness** (6-point semantic differential): evaluated the degree to which the stimulus was perceived as abstract versus concrete, defined by the anchors *"Very abstract"* (1) and *"Very concrete"* (6). Abstraction is a key design variable in neurofeedback, as highly representational visuals may induce semantic interpretation.
- **Aesthetic quality** (6-point Likert scale): captured overall visual appeal, with responses ranging from *"Aesthetically displeasing"* (1) to *"Aesthetically pleasing"* (6). This measure reflects aesthetic judgment and perceptual pleasantness.
- **Engagement** (6-point Likert scale): measured sustained visual interest, operationalized as perceived boredom. The scale ranged from *"Not boring at all"* (1) to *"Extremely boring"* (6), with higher values indicating lower engagement.
- **Cognitive load** (6-point Likert scale): assessed the perceived mental effort required to interpret and follow the visual feedback, using anchors from *"Very low cognitive load"* (1) to *"Very high cognitive load"* (6).

Together, these measures provide a multidimensional characterization of how stimuli are perceived in terms of clarity, complexity, abstraction, aesthetic value, engagement, and cognitive demand. This combination enables systematic evaluation of perceptual suitability and practical viability of generative visual feedback in neurofeedback contexts.

Visual Stimuli Figure 4.6 presents the seven visual stimuli evaluated in the user study. Five stimuli were generated using the proposed generative AI pipeline and represent different abstract visuals that meet NFB requirements (Flow, Galaxy Spiral, Neuron, Globe, Sunflower). Two additional stimuli (Ugly Thermo and Pleasing Thermo) were included as counter-examples, representing intentionally suboptimal traditionnal visual feedback designs.
All stimuli were presented as short animated sequences with identical duration and signal dynamics, ensuring that observed perceptual differences could be attributed primarily to visual design rather than temporal or interaction effects.

Figure 4.6: Overview of the seven visual stimuli evaluated in the user study. The first five visuals (Flow, Galaxy Spiral, Neuron, Globe, Sunflower) were generated using the proposed AI-based pipeline, while the last two (Ugly Thermo, Pleasing Thermo) serve as counter-examples.

Data Collection All responses were collected via an online survey platform. In addition to subjective ratings, basic interaction metadata (e.g., response time) were logged for quality control. The study is exploratory in nature; results are used to inform design iteration rather than to support clinical efficacy claims.

4.2.2. Results

Overview of Quantitative Ratings To provide a compact overview of perceptual performance across all criteria, a heatmap 4.7 representation was constructed with visual stimuli on the x-axis and evaluation dimensions on the y-axis. Cell values correspond to aggregated central tendencies (median ratings), allowing rapid comparison across stimuli and criteria.
This representation highlights overall performance trends and facilitates identification of stimuli that consistently perform well or poorly across multiple design objectives.

Engagement Analysis Sustained engagement 4.8 is a central motivation for using generative AI in neurofeedback, particularly for long clinical sessions. Engagement ratings were therefore analyzed in greater detail using violin plots.

Comparative Visual Profiles To compare overall perceptual profiles, radar (spider) plots were used to contrast aggregated ratings between two groups: AI-generated visuals and counter-example visuals. Radar plots provide an intuitive multidimensional overview, making trade-offs between design objectives immediately visible.

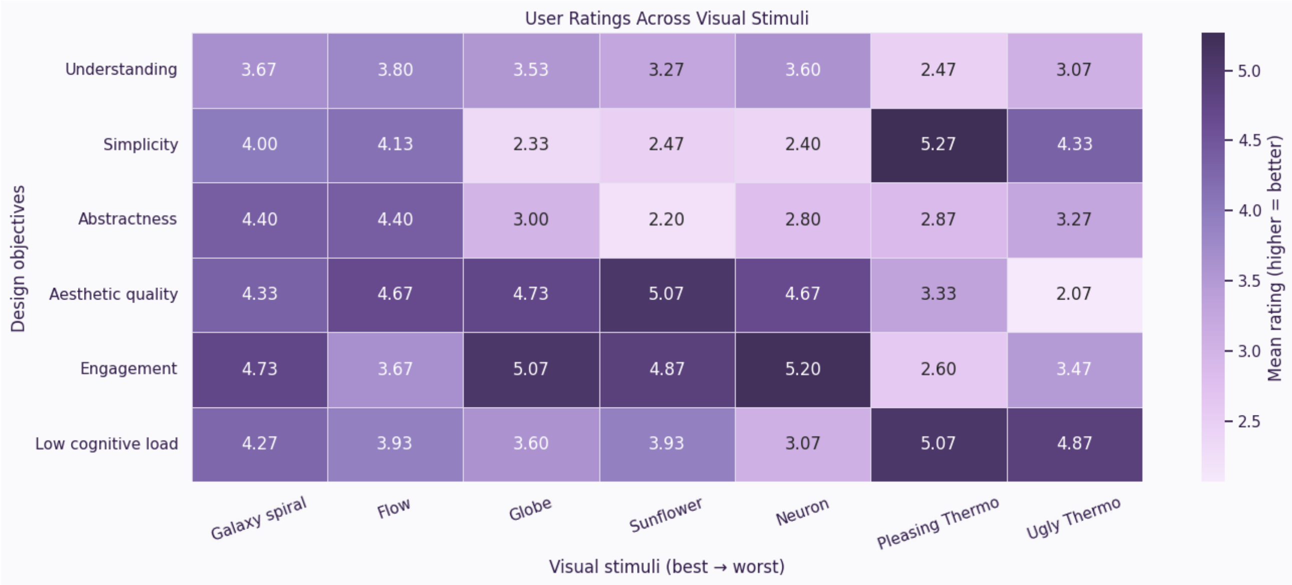

Figure 4.7: Heatmap summarizing median perceptual ratings across all visual stimuli and evaluation dimensions.

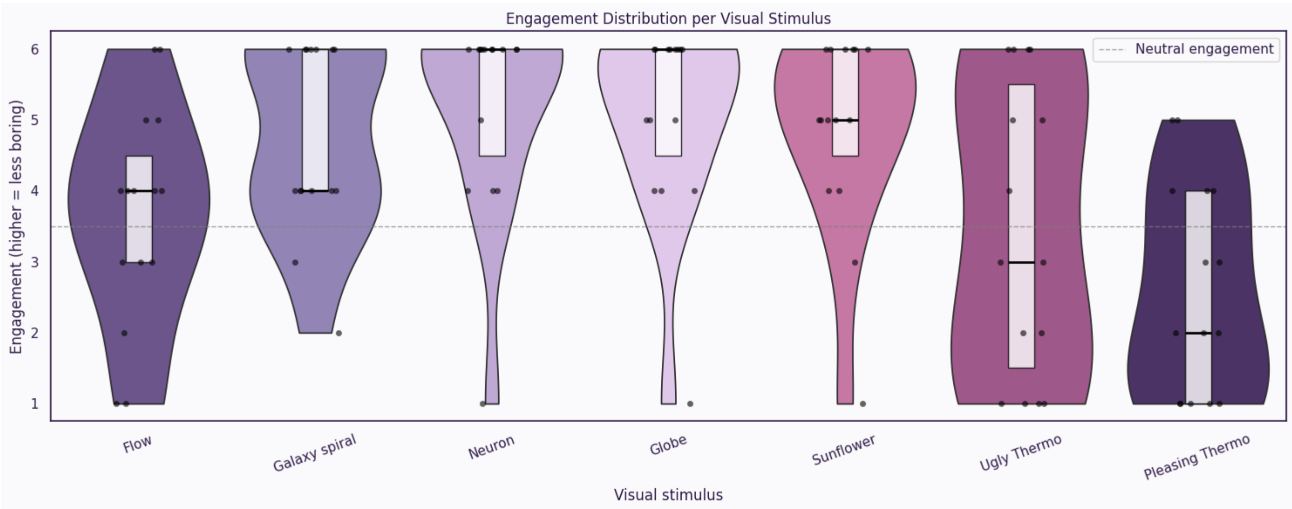

Figure 4.8: Distribution of engagement ratings per visual stimulus. Higher values indicate greater engagement (less boredom).

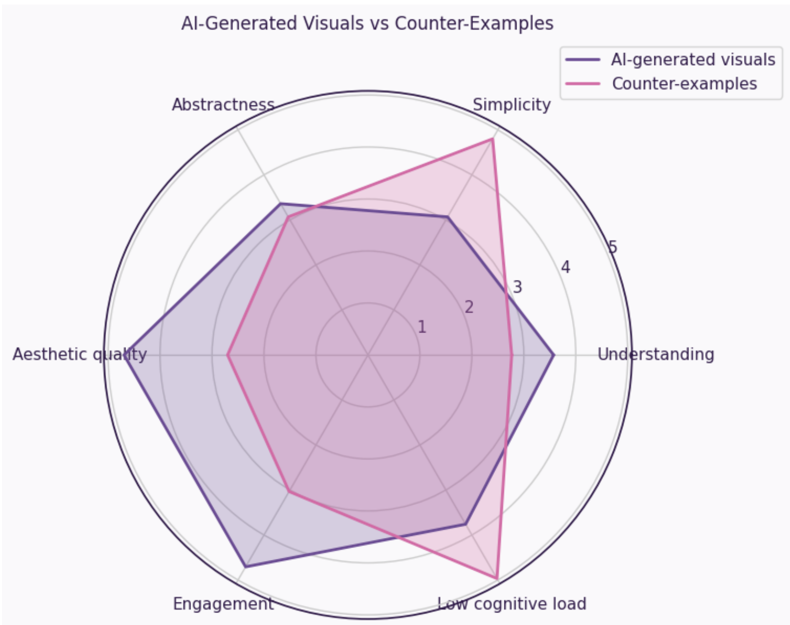

Figure 4.9: Radar plot comparing aggregated perceptual profiles of AI-generated visuals and counter-example stimuli.

5
Discussion

This chapter interprets the experimental findings and situates them within the broader design and technical choices of the project. Itfocuses on *why* specific approaches succeeded or failed, the trade-offs involved, and the implications for neurofeedback-oriented generative systems.

5.1. Model-by-Model Analysis

This section synthesizes qualitative observations and quantitative benchmarks (Table 4.1) to analyze the suitability of each evaluated generative model for neurofeedback visualization. Each model is assessed with respect to temporal coherence, controllability, training alignment, and compatibility with continuous signal-driven feedback.

Stable Video Diffusion (SVD) Stable Video Diffusion produced short clips with internally coherent motion; however, its applicability was severely constrained. The model achieved only 0.96 fps—the slowest among all evaluated systems—with the poorest temporal coherence metrics (LPIPS mean = 0.23, LPIPS max = 0.55; Table 4.1). These values, an order of magnitude higher than competing approaches, reflect substantial frame-to-frame perceptual variation despite the model's nominal video generation capability.
The model does not support text-based conditioning and relies on auxiliary parameters such as motion buckets and noise augmentation, which provide only indirect and coarse control over visual evolution (Control score = 0; Table 4.2). Generated motion was frequently dominated by camera-like effects (e.g., pans or zooms) that are inappropriate for abstract or symbolic feedback. Furthermore, clip duration is limited to approximately four seconds, and no mechanism exists for mapping a continuous control signal to visual state changes (Conditioning Feasibility score = 1; Table 4.2). These limitations rendered SVD unsuitable for neurofeedback use despite its nominal video coherence.

Matrix-Game and World Simulation Models Matrix-Game models operate as neural simulators trained exclusively on fixed game environments and conditioned on discrete control inputs. While they achieved moderate temporal coherence (LPIPS mean = 0.048, LPIPS max = 0.21; Table 4.1)—reflecting strong internal consistency due to deterministic physics simulation—their latent spaces encode environment-specific textures, layouts, and dynamics. As a result, they lack support for abstract, minimalist, or freely compositional visuals (Training Fit score = 0; Table 4.2).
The model's effective throughput of 0.81 fps (Table 4.1) combined with 16.69 GB peak VRAM usage indicates substantial computational cost. Conditioning is limited to discrete actions rather than continuous scalar inputs (Control score = 0, Conditioning Feasibility score = 0; Table 4.2), making it fundamentally incompatible with neurofeedback paradigms. Consequently, Matrix-Game models were excluded early from further experimentation.

LTX Video LTX Video demonstrated the strongest temporal coherence among video generation models (LPIPS mean = 0.0027, LPIPS max = 0.0156; Table 4.1) and relatively high generation speed (6.44 fps) compared to other video diffusion systems. However, this performance came at the cost of the highest memory footprint (17.5 GB peak VRAM; Table 4.1). More critically, its training on large-scale photorealistic internet video induces a strong bias toward natural scenes, lighting effects, depth cues, and cinematic motion (Training Fit score = 1; Table 4.2). When prompted for abstract or graphic visuals, outputs frequently exhibited texture leakage, unintended shading, and unstable geometric structure. Although coherence was high within realistic domains (Temporal Coherence score = 3; Table 4.2), the lack of frame-level control (Control score = 2; Table 4.2) and the difficulty of enforcing minimalist aesthetics made LTX Video ill-suited for neurofeedback visualization despite its favorable performance metrics.

StreamDiffusion (SD-Turbo) StreamDiffusion achieved the highest effective frame rate (13.53 fps) and the lowest VRAM usage (2.71 GB) among all evaluated systems (Table 4.1), confirming the effectiveness of pipeline-level acceleration (Speed score = 3; Table 4.2). However, the absence of explicit temporal modeling resulted in severe frame-to-frame inconsistency. This was quantified by elevated perceptual variation (LPIPS mean = 0.032, LPIPS max = 0.10; Table 4.1)—approximately 12× higher mean LPIPS than LTX Video—manifesting visually as flickering, color drift, and shape instability. While suitable for interactive visual effects or rapid prototyping, StreamDiffusion failed to meet the temporal smoothness requirements necessary for sustained neurofeedback sessions (Temporal Coherence score = 2; Table 4.2), despite offering moderate controllability (Control score = 2; Table 4.2).

Stable Diffusion XL (SDXL) Although SDXL achieved only 3.03 fps—too slow for direct real-time generation (Speed score = 2; Table 4.2)—it consistently provided the highest visual quality (Quality score = 2; Table 4.2) and strongest controllability (Control score = 2; Table 4.2). Its temporal coherence metrics (LPIPS mean = 0.058, LPIPS max = 0.095; Table 4.1) were moderate, reflecting frame-to-frame variation inherent to image-based diffusion without explicit temporal priors. The `img2img` pipeline enabled deterministic, reproducible frame-to-frame evolution when using low denoising strengths (0.20–0.30), allowing precise adherence to design constraints such as centered composition, background purity, and gradual evolution (Conditioning Feasibility score = 3; Table 4.2). With moderate VRAM usage (7.69 GB; Table 4.1) and strong training alignment (Training Fit score = 2–3; Table 4.2), SDXL's image–text training distribution supports abstract, non-photoreal, and graphic visuals without environmental constraints, making it uniquely well aligned with neurofeedback requirements.

LoRA Fine-Tuning Attempts Low-Rank Adaptation (LoRA) was explored as a means of enforcing stylistic consistency across generated visuals. In practice, LoRA training was unsuccessful, likely due to limited dataset size and high stylistic variability among target visuals. As a result, the final system relies on explicit prompt engineering and controlled `img2img` parameters rather than learned style adapters.

5.2. Synthesis of Performance Results

Quantitative benchmarks (Table 4.1) highlight the inherent trade-offs between speed, temporal coherence, and resource usage across models. StreamDiffusion excels in speed (13.53 fps, 2.71 GB VRAM) but lacks stability (LPIPS mean = 0.032); LTX Video achieves excellent coherence (LPIPS mean = 0.0027) at high memory cost (17.5 GB) and limited control; SVD and Matrix-Game models fail to meet multiple core requirements simultaneously. SDXL, while not the fastest (3.03 fps), occupies a balanced position with moderate memory usage (7.69 GB), offering strong visual quality, moderate temporal coherence (LPIPS mean = 0.058), and high controllability. These findings motivate the ordinal synthesis presented in Table 4.2, where SDXL emerges as the most suitable foundation for neurofeedback visualization.

5.3. Model Selection and Generation Strategy

A central conclusion of this work is that no existing generative model simultaneously satisfies real-time generation (24+ fps), high visual fidelity, fine-grained control, and stable temporal coherence—as evidenced by the complementary strengths and weaknesses documented in Tables 4.1 and 4.2. Models optimized for speed sacrifice stability or controllability, while high-quality diffusion models remain computationally expensive.
This constraint motivated the selection of **Stable Diffusion XL (SDXL)** as the core generative model despite its 3.03 fps generation rate (Table 4.1). Its strengths—visual quality, compositional control, and predictable behavior under `img2img`—align closely with neurofeedback design requirements (highest scores for Control and Conditioning Feasibility; Table 4.2).
To resolve the real-time constraint, a hybrid strategy was adopted: *offline keyframe generation followed by real-time playback*. High-quality frames are generated without time pressure, while real-time constraints are met through lightweight decoding and display. This approach decouples visual quality from temporal requirements and represents a pragmatic resolution to current limitations of diffusion-based video generation.

5.4. Temporal Continuity and Latent Interpolation

The choice of interpolation method emerged as a critical determinant of perceptual quality. Quantitative evaluation revealed substantial differences in latent space behavior across interpolation techniques (Table 4.3). While perceptual similarity metrics (LPIPS) were comparable across methods—LERP and SLERP both achieved mean values of 0.002—**norm deviation** exhibited dramatic variation: SLERP preserved latent magnitudes with only 0.008% deviation, compared to 2% for LERP and 1.5% for cosine interpolation (Table 4.3).

Why Norm Preservation Matters Latent variables in diffusion models are approximately Gaussian distributed with expected norms on the order of $\sqrt{d}$ for dimensionality d. Linear interpolation (LERP) between two latent samples reduces the norm at intermediate points—causing variance collapse of approximately 50% at the midpoint—resulting in off-manifold inputs for the VAE decoder. This statistical mismatch manifests perceptually as reduced sharpness, color desaturation, and luminance shifts in decoded images.
Spherical Linear Interpolation (SLERP) addresses this by interpolating along the geodesic of the hypersphere defined by the normalized latent directions, rather than along a straight Euclidean path. By preserving the expected latent norm throughout the interpolation trajectory, SLERP maintains consistency with the diffusion model's training distribution, yielding intermediate frames that remain on-manifold.

The 250-fold reduction in norm deviation (from 2% to 0.008%; Table 4.3) directly translates to improved perceptual stability. While LPIPS captures local perceptual similarity between adjacent frames, norm preservation ensures that intermediate frames maintain the statistical properties expected by the decoder—preventing the gradual quality degradation that would accumulate across the 35 interpolated frames between keyframes.

Implications for Neurofeedback Given that interpolated frames constitute the majority of displayed content in neurofeedback sessions (35 of every 36 frames in our configuration), SLERP's superior norm preservation substantially improves perceptual continuity and user comfort. The method's computational overhead—consisting of vector normalization, dot product computation, and trigonometric evaluation—adds only 87.2 seconds to the offline preparation of a 30-second sequence (Table 4.4), which is negligible relative to the total preparation time of 172.4 seconds.
The selected configuration of 21 keyframes with 35 interpolations per segment reflects a deliberate balance between smooth motion perception (achieved at 24 fps) and offline preparation cost. While denser interpolation would further improve smoothness, it scales linearly in preparation time (single keyframe: 1.15s; 21 keyframes: 24.1s; Table 4.4), motivating adaptive strategies for extended sessions where computational budget becomes constrained.

5.5. Prompt Engineering and VLM-Based Automation

Manual prompt design proved insufficient for generating long, coherent evolution sequences under strict aesthetic constraints. Integrating a vision–language model (Qwen2.5-VL) for prompt automation enabled systematic extraction of style attributes and controlled prompt interpolation, significantly reducing authoring effort while improving consistency. The VLM prompt generation stage required an average of 67 seconds per sequence (Table 4.4)—approximately 39% of the total offline preparation time—but eliminated the need for manual prompt crafting across multiple keyframes, making the approach viable for generating diverse stimulus libraries.

5.6. System Performance and Practical Constraints

From a systems perspective, the hybrid pipeline meets neurofeedback latency requirements comfortably. The complete offline generation process for a 30-second sequence (721 frames at 24 fps) requires 172.4 seconds on average (Table 4.4), dominated by SLERP interpolation (87.2s) and keyframe generation (24.1s for 21 keyframes), with VAE encoding/decoding contributing minimal overhead (0.03s and 0.05s per frame, respectively).
During real-time playback, GPU resources are not required, and end-to-end latency remains at 19–23 ms (Table 4.5)—well below the $\sim$100 ms threshold typically considered acceptable for neurofeedback applications. CPU usage stays at 5–12%, with RAM consumption of 200–400 MB (Table 4.5). Disk I/O from NVMe SSD (2–6 ms latency) constitutes the dominant source of playback latency, which could be eliminated entirely through RAM caching.
Storage requirements are modest: a single 30-second sequence occupies approximately 110–120 MB (70–170 MB for PNG frames, <0.5 MB for metadata, 8–15 MB for H.264-encoded video). A standard 1 TB NVMe SSD can therefore store 8,000–9,000 unique sequences, corresponding to 65–75 hours of non-repeating stimuli. For long-duration sessions, dense frame generation becomes computationally expensive. A reduced frame-density strategy—combined with frame reuse when the signal revisits similar ranges—offers a viable compromise, significantly reducing preparation time at the cost of potential visual repetition. This trade-off highlights the tension between scalability and variability in generative feedback systems.

5.7. User Testing and Perceptual Implications

User testing results (Figures 4.7, 4.8, 4.9) support the core motivation of the project: enhancing engagement in neurofeedback through generative visuals. Violin plot analysis (Figure 4.8) revealed that AI-generated stimuli achieved higher median engagement scores with lower inter-participant variability compared to counter-example visuals. This concentration of ratings in the upper range suggests not only higher engagement but also greater perceptual consistency across users.
The heatmap overview (Figure 4.7) demonstrates that AI-generated stimuli (Flow, Galaxy Spiral, Neuron, Globe, Sunflower) consistently outperformed counter-examples (Ugly Thermo, Pleasing Thermo) across multiple evaluation dimensions. Spider plot comparisons (Figure 4.9) further indicate that AI-generated visuals achieve a more favorable balance between engagement, aesthetic quality, and cognitive load. Counter-example stimuli exhibited pronounced weaknesses in engagement and perceptual coherence, reinforcing the importance of controlled generative design rather than ad hoc visual complexity.
While these findings remain exploratory, they provide initial evidence that generative, evolving visuals may better sustain attention during extended neurofeedback sessions—a key requirement for clinical viability.

5.8. Design Trade-offs and Limitations

Despite its strengths, the proposed approach entails inherent trade-offs. Pre-generation limits immediate adaptation to unexpected neural patterns, and the system cannot respond dynamically to unforeseen user behavior during a session. The offline preparation time of 172.4 seconds per 30-second sequence (Table 4.4) necessitates advance planning and constrains real-time personalization. Expressive visual evolution must be carefully balanced against cognitive load: increased complexity improves perceptual clarity but risks attentional overload.

Several technical limitations remain. LoRA-based fine-tuning for style consistency was unsuccessful with the available dataset, and SDXL's inductive biases occasionally surfaced despite careful prompting. Most importantly, validation has not yet extended to real EEG-driven neurofeedback sessions or clinical populations, and long-term learning effects remain unknown.

5.9. Implications and Future Directions

This work demonstrates the feasibility of integrating generative AI into neurofeedback visualization pipelines and suggests a shift from static, parameter-driven displays toward evolving, personalized feedback. The hybrid generation paradigm explored here—combining offline high-quality generation (Table 4.4) with real-time lightweight playback (Table 4.5)—generalizes beyond neurofeedback and may inform adaptive visual systems in biofeedback, meditation, and interactive media.

Future work should prioritize clinical validation with real EEG signals, longitudinal engagement studies to assess sustained effectiveness, development of scalable stimulus libraries, and investigation of hybrid online–offline generation schemes that balance preparation cost with adaptive flexibility. More broadly, this project highlights how careful system design—rather than raw model capability alone—determines whether generative AI can meet the stringent constraints of real-time, human-centered applications.

6 Conclusion

This work addressed a fundamental challenge in neurofeedback visualization: generating high-quality, controllable visual feedback at real-time frame rates. Systematic evaluation of five generative models (Table 4.1) revealed that no existing architecture simultaneously satisfies the requirements of speed (>24 fps), temporal coherence (LPIPS < 0.01), fine-grained control, and abstract aesthetic compatibility. Models optimized for speed (StreamDiffusion: 13.53 fps) exhibited severe frame instability, while temporally coherent models (LTX Video: LPIPS = 0.0027) remained computationally expensive and biased toward photorealism.

To resolve these constraints, we developed a hybrid pipeline combining offline SDXL keyframe generation, Qwen2.5-VL prompt automation, and SLERP latent interpolation for real-time playback. SLERP achieved 250-fold better norm preservation than linear interpolation (0.008% vs 2% deviation; Table 4.3), ensuring perceptual stability across the 35 interpolated frames between keyframes. The complete system delivers 24 fps playback with 19–23 ms latency (Table 4.5) without requiring GPU resources, while maintaining aesthetic quality and compositional control.

Preliminary user testing with 18 participants demonstrated that AI-generated visuals achieved higher engagement and more favorable perceptual profiles than traditional neurofeedback displays (Figures 4.8, 4.9). However, clinical efficacy remains to be validated through integration with EEG systems and controlled longitudinal studies with target populations.

The core contribution of this work is demonstrating that carefully designed hybrid architectures—rather than raw generative model performance—enable deployment of diffusion models in domains with strict real-time and perceptual requirements. The proposed framework is applicable beyond neurofeedback to biofeedback, meditation interfaces, and therapeutic visualization, providing a methodological foundation as generative models continue to evolve.

References

[1] D. Corydon Hammond. “Neurofeedback: A comprehensive review”. In: *Journal of Neurotherapy* 10 (2005), pp. 79–90. DOI: `10.1300/J184v10n01_04`.

[2] Ranganatha Sitaram, Tomas Ros, Leigh Stoeckel, et al. “Closed-loop brain training: the science of neurofeedback”. In: *Nature Reviews Neuroscience* 18 (2017), pp. 86–100. DOI: `10.1038/nrn.2016.164`.

[3] Joe Kamiya. “Conscious control of brain waves”. In: *Psychology Today* 1 (1968), pp. 56–60.

[4] R.A Kovalesky M.B Sterman S.J Goodman. “Effects of sensorimotor EEG feedback training on seizure susceptibility in the rhesus monkey”. In: *Annals of Neurology* 1 (1970), pp. 455–462.

[5] Byung In Han et al. “Tinnitus: Characteristics, Causes, Mechanisms, and Treatments”. In: *Journal of Clinical Neurology* 5.1 (2009), pp. 11–19. DOI: `10.3988/jcn.2009.5.1.11`.

[6] Christian M. Jarach, Alessandra Lugo, Martina Scala, et al. “Global Prevalence and Incidence of Tinnitus: A Systematic Review and Meta-analysis”. In: *JAMA Neurology* 79 (2022), pp. 888–900. DOI: `10.1001/jamaneurol.2022.2189`.

[7] Ruchika Biswas, Alessandra Lugo, Michael A. Akeroyd, et al. “Tinnitus prevalence in Europe: A multi-country cross-sectional population study”. In: *The Lancet Regional Health – Europe* 12 (2022), p. 100250. DOI: `10.1016/j.lanepe.2021.100250`.

[8] David Güntensperger et al. “Neurofeedback for tinnitus treatment—review and meta-analysis”. In: *Frontiers in Neuroscience* 13 (2019), p. 386. DOI: `10.3389/fnins.2019.00386`.

[9] Roland Bauer and Sidney Fels. “Designing feedback for neurofeedback learning”. In: *Frontiers in Human Neuroscience* 9 (2015), p. 654. DOI: `10.3389/fnhum.2015.00654`.

[10] Cai et al. “Toward Design Principles for Neurofeedback Stimuli: A Design Evaluation Framework Based on Medium and Message”. In: (2025).

[11] Robin Rombach et al. “High-Resolution Image Synthesis with Latent Diffusion Models”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2022), pp. 10684–10695.

[12] Dustin Podell et al. “SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis”. In: *arXiv preprint arXiv:2307.01952* (2023).

[13] Jinze Bai et al. “Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond”. In: *arXiv preprint arXiv:2308.12966* (2023).

[14] Jonathan Ho, Tim Salimans, and Alexey Gritsenko. “Video Diffusion Models”. In: *Advances in Neural Information Processing Systems* (2022).

[15] Andreas Blattmann, Robin Rombach, and Huan Ling. “Align Your Latents: High-Resolution Video Synthesis with Latent Diffusion Models”. In: *arXiv preprint arXiv:2304.08818* (2023).

[16] Dennis Küstner et al. “The power of generative marketing: Can generative AI create superhuman visual marketing content?” In: *ScienceDirect* (2024). Available at: `https://www.sciencedirect.com/science/article/pii/S0167811624000843`.

[17] Tijmen Jansen et al. “Automated Alignment: Guiding Visual Generative AI for Brand Building and Customer Engagement”. In: ResearchGate, 2023. DOI: `10.13140/RG.2.2.11234.56789`.

[18] Tobias Kleinjung, Martin Meyer, and Patrick Neff. “Neurofeedback for tinnitus treatment: an innovative method with promising potential”. In: *Brain Communications* 5.4 (2023), fcad209. DOI: `10.1093/braincomms/fcad209`.

[19] IEEE Conference Publication. “Designing Feedback Stimuli in Neurofeedback: Preliminary Requirements from Experts and Users”. In: *IEEE Conference Proceedings*. Available at: `https://ieeexplore.ieee.org/document/11058781/`. IEEE, 2024.

[20] Stefanie Enriquez-Geppert et al. “Systematic review and meta-analysis of the relationships between real-time neurofeedback training parameters and acquisition of neural modulation”. In: *Frontiers in Human Neuroscience* 19 (2025), p. 1652607. DOI: `10.3389/fnhum.2025.1652607`.

[21] Emmanuel Dessy, Martine Van Puyvelde, Olivier Mairesse, et al. “Neurofeedback of Alpha Activity on Memory in Healthy Participants: A Systematic Review and Meta-Analysis”. In: *Frontiers in Human Neuroscience* 14 (2020), p. 562360. DOI: `10.3389/fnhum.2020.562360`.

[22] Dominik Güntensperger et al. “Investigating the Efficacy of an Individualized Alpha/Delta Neurofeedback Protocol in the Treatment of Chronic Tinnitus”. In: *Neural Plasticity* 2019 (2019), p. 3540898. DOI: `10.1155/2019/3540898`.

[23] Robert Bauer and Alireza Gharabaghi. “Estimating cognitive load during self-regulation of brain activity and neurofeedback with therapeutic brain-computer interfaces”. In: *Frontiers in Behavioral Neuroscience* 9 (2015), p. 21. DOI: 10.3389/fnbeh.2015.00021.

[24] Matthias Witte et al. “Predictors of neurofeedback training outcome: A systematic review”. In: *NeuroImage: Clinical* 27 (2020), p. 102301. DOI: 10.1016/j.nicl.2020.102301.

[25] Oded Alkoby et al. “Can we predict who will respond to neurofeedback? A review of the inefficacy problem and existing predictors for successful EEG neurofeedback learning”. In: *Neuroscience* 378 (2018), pp. 155–164. DOI: 10.1016/j.neuroscience.2016.12.050.

[26] Markus Plank et al. “Combining EEG and eye tracking: identification, characterization, and correction of eye movement artifacts in electroencephalographic data”. In: *Frontiers in Human Neuroscience* 6 (2012), p. 278. DOI: 10.3389/fnhum.2012.00278.

[27] Yoav HaCohen et al. “LTX-Video: Realtime Video Latent Diffusion”. In: *arXiv preprint arXiv:2501.00103* (2024).

[28] Andreas Blattmann et al. “Stable Video Diffusion: Scaling Latent Video Diffusion Models to Large Datasets”. In: *arXiv preprint arXiv:2311.15127* (2023).

[29] Xianglong He et al. “Matrix-Game 2.0: An Open-Source, Real-Time, and Streaming Interactive World Model”. In: *arXiv preprint arXiv:2508.13009* (2025).

[30] Akio Kodaira et al. “StreamDiffusion: A Pipeline-Level Solution for Real-Time Interactive Generation”. In: *arXiv preprint arXiv:2312.12491* (2023).

[31] Axel Sauer et al. “Adversarial Diffusion Distillation”. In: *arXiv preprint arXiv:2311.17042* (2023). Technical report for SDXL-Turbo.

[32] Edward J Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models”. In: *International Conference on Learning Representations*. 2022.

[33] Shuai Bai et al. “Qwen2.5-VL Technical Report”. In: *arXiv preprint arXiv:2502.13923* (2025).

[34] Qwen Team. *Qwen2.5: A Party of Foundation Models*. https://qwenlm.github.io/blog/qwen2.5/. 2024.

[35] Pan Lu et al. “MathVista: Evaluating Mathematical Reasoning of Foundation Models in Visual Contexts”. In: *arXiv preprint arXiv:2310.02255* (2024).

[36] Minesh Mathew, Dimosthenis Karatzas, and CV Jawahar. “DocVQA: A Dataset for VQA on Document Images”. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* (2021), pp. 2200–2209.

[37] Richard Zhang et al. “The Unreasonable Effectiveness of Deep Features as a Perceptual Metric”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2018, pp. 586–595.

[38] Ken Shoemake. “Animating Rotation with Quaternion Curves”. In: *ACM SIGGRAPH Computer Graphics*. Vol. 19. 3. ACM. 1985, pp. 245–254.

[39] Tom White. “Sampling Generative Networks”. In: *arXiv preprint arXiv:1609.04468* (2016).

A
Appendix

A.1. Code Repository Structure

Complete project code available at: `https://github.com/coralieb7/semester-project`
Key Files:

```
scripts/2_generate_images.py: Main generation pipeline
scripts/3_create_video.py: Video assembly with signal functions
src/vlm/prompt_generator.py: VLM-based prompt engineering
src/sdxl/latent_slerp_interpolator.py: SLERP implementation
src/video/video_generator.py: FFmpeg video creation
```

A.1.1. Pipeline

- Reference Image Selection: Choose image exemplifying desired final aesthetic.
  - Example: Abstract purple sunflower-like form with translucent petals on white background
- Image Analysis (Qwen2.5-VL):

```
analysis = analyzer.analyze_image(
            image_path='data/input/reference.png',
            analysis_type='brief' # or 'detailed', 'technical'
        )
```

  Output example:

```
"A large, centered form resembling an abstract sunflower is composed of multiple soft, translucent layers in varying
    shades of white, pink, and purple. Positioned centrally on a pristine white background."
```

- Evolution Goal Expansion (New Feature):
  - User provides brief evolution description: ”the dot becomes a beautiful sunflower”
  - VLM expands into detailed, vivid description:

```
expanded_goal = generator.expand_evolution_goal(
    evolution_goal="the␣dot␣becomes␣a␣beautiful␣sunflower",
    style_guidelines={
        "background": "white␣background",
        "composition": "centered,␣balanced",
        "color_palette": "purple,␣pink,␣white",
        "style": "minimalist␣graphic␣design"
    }
)
```

  Output:

```

"A large, centered sunflower with soft, translucent purple and pink petals radiating from a deep purple core,
    positioned centrally on a pristine white background. The petals have an abstract, flowing quality with subtle
     gradients from light pink to deep purple. The composition is balanced and minimalist with clean lines and
    modern aesthetic."
```

- Evolutionary Prompt Sequence Generation:

```
prompts = generator.generate_evolutionary_sequence(
    image_path='data/input/reference.png',
    evolution_description=expanded_goal,
    num_prompts=20,
    prompts_type='basic'  # or 'only', 'few\_shot', 'matrix'
)
```

A.1.2. Example prompt sequences

Example: "Dot to Sunflower" Evolution

Step 1 (signal=0.0):

```
{ "step": 1,
  "signal": 0.0,
  "prompt": "A large, centered form resembling an abstract sunflower is composed of multiple soft, translucent layers in varying shades of white, pink, and purple. Positioned centrally on a pristine white background.",
  "change": "Initial state matching analyzed image"
}
```

Step 10 (signal=0.47):

```
{ "step": 10,
  "signal": 0.47,
  "prompt": "A large, centered form resembling an abstract sunflower with developing petal structure in soft shades of white, pink, and purple. Positioned centrally on white background with subtle gradients and increasingly defined petal edges. Balanced composition with emerging organic complexity.",
  "change": "Petal definition increasing, gradients developing"
}
```

Step 20 (signal=1.0):

```
{ "step": 20,
  "signal": 1.0,
  "prompt": "A large, centered form resembling an abstract sunflower is composed of multiple soft, translucent layers in varying shades of white, pink, and purple. Positioned centrally on pristine white background with subtle gradient from light purple to white, featuring clearly defined petal-like shapes with fluid organic forms and gentle motion implied by subtle gradients and overlapping edges, with enhanced light and shadow for dynamic tranquility, including clean geometric outlines for modern simplicity, with focus on delicate interplay between light and shadow for visually striking and tranquil visual experience, now incorporating detailed geometric patterns and soft gradients  throughout, with subtle suggestion of central core to enhance sunflower appearance, now with hint of intricate seed-like patterns in center to further define sunflower theme, now with more pronounced central core and additional petal-like structures around edges to create fuller, more complex form, now with refined central core, additional petal-like structures, and subtle shading for depth, now with more defined center and subtle highlights to enhance overall tranquility and elegance, now with polished and refined look, emphasizing clean lines and interplay between light and shadow for serene and elegant visual experience.",
  "change": "Fully realized sunflower form with maximum detail and complexity"
}
```

A.1.3. Implementation

Generate evolutionary image sequence

```
python scripts/2_generate_images.py \
--initial-image data/input/reference.png \
--evolution "the dot becomes a beautiful sunflower" \
--prompts-type basic \
--num-prompts 20 \
--num-keyframes 20 \
--interpolate \
--interpolations 35 \
--interpolator latent_slerp \
--strength 0.30 \
--guidance 15.0 \
--steps 50 \
--seed 42
```

Create video from generated frames

```
python scripts/3_create_video.py \
--project generation_[timestamp] \
--signal sine \
--frequency 1.0 \
--duration 30 \
--fps 24
```

A.2. User Testing Materials

User testing survey (Qualtrics)